

Zain Khan

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EDUCATION

University of Waterloo

Waterloo, ON

Bachelor of Applied Science, Computer Engineering

Sep. 2025 – Apr. 2030

– Awarded over **\$40,000** in merit-based scholarships, including the **University of Waterloo Presidential Scholarship**.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, Ruby, C/C++, Java, SQL, HTML/CSS

Frameworks: React, Redux, Next.js, Node.js, Express, Django, FastAPI, Three.js, TensorFlow, OpenCV, Tailwind CSS

Infrastructure: PostgreSQL, MySQL, MongoDB, Supabase, AWS, Docker, Linux, Git, Vercel

Protocols: HTTP/REST, GraphQL, gRPC, WebRTC, WebSockets, OAuth 2.0

EXPERIENCE

Leap Tools

Toronto, ON

Software Development Intern, Bespoke Team

May 2026 – Aug. 2026

- Built and shipped Surface Designer end-to-end (TypeScript, React/Redux, Next.js) on the Roomvo visualizer used in **80+ countries**: moved tile-pattern generation from a server-side C++ render service into the browser via a parametric format and one generic lattice-expansion algorithm, enabling live in-browser pattern editing with zero per-render round-trips; shipped **19** production patterns.
- Delivered commerce features across **6** more production apps, integrating Python/Django microservices over HTTP and gRPC on PostgreSQL and AWS: CORS-enabled REST APIs serving visitor data to external vendor storefronts, **10**-event PostHog analytics, and a full-stack persistence slice with a Postgres/Kysely migration.
- Authored **402** tests including the app's first Playwright E2E suite (**21** specs); diagnosed and fixed an SSR-hydration flake class that was breaking E2E runs.

WATonomous

Waterloo, ON

Robotics Software Engineer, EVE Autonomy

Jan. 2026 – Present

- Developed concurrent autonomous robotics software in C++ on ROS 2 Humble, coordinating **5+ real-time nodes** across perception, planning, and control subsystems for an autonomous delivery platform.
- Implemented costmaps, A* path planning, and Pure Pursuit control, achieving stable point-to-point navigation in cluttered indoor environments with dynamic obstacle avoidance.

Kasperia

Remote

Software Developer (Contract), Frontend & Data

Jun. 2023 – Sep. 2025

- Architected reusable React components with modular CSS across the client-facing web application, improving page load performance by **30%** and reducing maintenance overhead by **25%**.
- Built automated data pipelines using JavaScript, SQL, and REST APIs to consolidate reporting workflows, cutting manual reporting time by **40%** and powering real-time KPI dashboards.

PROJECTS

otto  | *Next.js, TypeScript, Python, LiveKit WebRTC, Deepgram, Gemini 2.5 Flash, Supabase*

- Shipped an AI voice agent that turns natural speech into automated actions across GitHub, Gmail, and Google Calendar; awarded **1st Place** — **LiveKit Track** at NexHacks.
- Engineered a LiveKit WebRTC pipeline with Deepgram VAD turn detection and Gemini 2.5 Flash Realtime native audio inference, achieving **sub-second** end-to-end voice-to-action latency.
- Integrated OAuth 2.0 auto-refresh via Supabase Auth and compressed agent context with TTC Bear-1, reducing per-request token usage from **4000** → **1800 (55%)**.

toothfairy  | *Next.js, TypeScript, FastAPI, Python, TensorFlow, OpenCV, Gemini 2.5 Flash, Moorcheh AI*

- Built a production-grade dental X-ray segmentation pipeline using a U-Net (TensorFlow/Keras) with OpenCV connected component analysis and morphological post-processing, isolating **28+ individual teeth** from panoramic radiographs.
- Integrated Moorcheh AI for per-patient RAG with isolated memory namespaces and semantic search across visits, enabling longitudinal retrieval for clinical history and diagnostic correlation.
- Orchestrated **3 AI copilots** using Gemini 2.5 Flash for structured JSON extraction and Gemini Vision for pathology detection with confidence scoring across **10 dental conditions**.

extgen | *React, TypeScript, Chrome Extensions, FastAPI, Python, Agentverse, uAgents, Gemini, WebSockets*

- Built a Chrome side panel that turns natural-language requests into installable Manifest V3 extensions, generating content scripts, CSS, validated project files, ZIP packages, and load-unpacked instructions.
- Implemented Edit DOM mode with Cmd/Ctrl-hover element selection, ordered edit chips, live DOM operations, and one-click export from temporary page edits into a permanent extension.
- Designed an Agentverse/uAgents orchestration pipeline with Architect, RAG, Codegen, Validator, and Packager roles, combining deterministic intent templates with LLM generation and per-site DOM bootstrap retrieval.